CS23334 Fundamentals of Data Science Lab

Experiment 6

Time Series Analysis of Website Traffic

Objective: To analyze the time series data of website traffic to identify trends, seasonality, and anomalies.

Dataset: A dataset containing daily website traffic data with columns such as `Date`, `Page_Views`, `Unique_Visitors`, and `Bounce_Rate`.

Methodology:

1. **Load and Clean Data:** Import the dataset and parse the Date column as a datetime object. Handle missing values by interpolation.
2. **Time Series Decomposition:** Decompose the time series data of Page_Views to identify trend, seasonality, and residuals.
3. **Visualization:** Plot the time series data for Page_Views, Unique_Visitors, and Bounce_Rate. Use moving averages to smooth the time series and highlight trends. Create seasonal plots to visualize patterns over time.
4. **Anomaly Detection:** Identify and visualize anomalies in the time series data using statistical methods such as the Z-score.

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

from statsmodels.tsa.seasonal import seasonal_decompose

# Load dataset
data = pd.read_csv('website_traffic.csv', parse_dates=['Date'], index_col='Date')

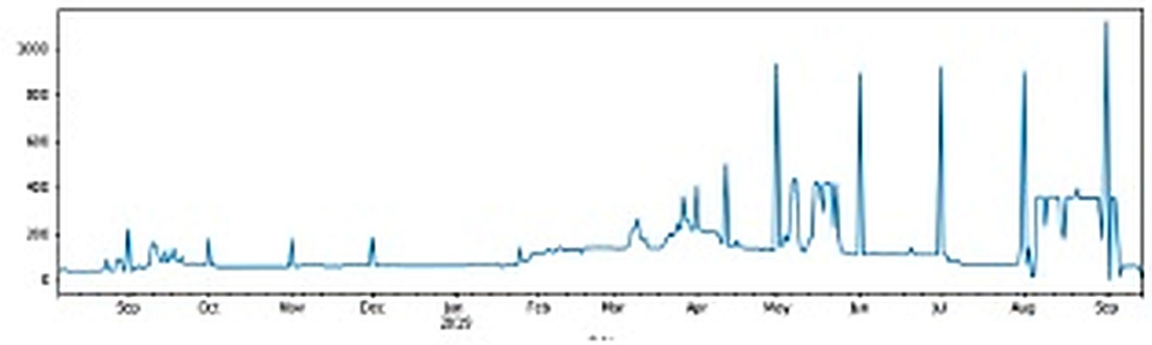
# Time Series Decomposition
decomposition = seasonal_decompose(data['Page_Views'], model='additive')
decomposition.plot()
plt.show()

# Moving Average
data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()
data[['Page_Views', 'Page_Views_MA']].plot()
plt.title('Page Views with Moving Average')
plt.show()

# Anomaly Detection
data['Page_Views_Z'] = (data['Page_Views'] - data['Page_Views'].mean()) / data['Page_Views'].std(
anomalies = data[data['Page_Views_Z'].abs() > 3]

plt.plot(data.index, data['Page_Views'], label='Page Views')
plt.scatter(anomalies.index, anomalies['Page_Views'], color='red', label='Anomalies')
plt.legend()
plt.show()
```

OUTPUT:



Expected Outcomes:

- Clear visualization of trends, seasonality, and anomalies in website traffic.
- Understanding of the underlying patterns in user behavior over time.

Result:

Time series decomposition of website traffic data successfully extracted trend, seasonality, and residual components. An upward trend in page views was observed with weekly seasonality, and anomalies were identified as significant spikes during promotional periods.